Subspaces and Closure

Sets Closed Under Operations

Defn. *A set S is* ***closed*** *under some operation if applying that operation to elements of S always produces an element of S.*

For example, the integers are closed under addition: adding two integers always produces an integer. The integers are also closed under subtraction and multiplication, but not division.

Sets Closed Under Operations

The set of positive real numbers is not closed under subtraction: for example, $2 - \pi$ is not positive. The set is closed under addition, multiplication, division, and exponentiation.

A Span is Closed

Fact. *The set of solutions to the homogeneous equation $Ax = \mathbf{0}$ is closed under both addition and scalar multiplication.*

In other word, if you add two solutions of the homogenous equation then the result is still a solution; similarly, if you scale a solution then the result is still a solution.

Subspaces

Defn. *A **subspace** of $\mathbb{R}^n$ is a subset such that*

(0) S contains the zero vector;

(1) S is closed under addition;

(2) S is closed under scalar multiplication.

Subspaces of $\mathbb{R}^2$

A line through the origin is a subspace of $\mathbb{R}^2$.

Not a Subspace

Consider the set of points (x, y) in $\mathbb{R}^2$ such that $|x| = |y|$. This is not a subspace: not closed under addition.

Consider the set of points (x, y) in $\mathbb{R}^2$ such that $x, y \geq 0$. This is not a subspace: not closed under scalar multiplication.

The Two Trivial Subspaces

Fact. *The set containing just the zero-vector is always a subspace.*
The whole space is always a subspace of itself.

Why?

Subspaces of $\mathbb{R}^3$

Fact. *Every subspace of* $\mathbb{R}^3$ *is either* $\{0\}$, *a line through the origin, a plane through the origin, or the space itself.*

Spans are Subspaces

Fact. *If S is a set of vectors, then $\mathit{Span}\, S$ is a subspace.*

Summary

A set is closed under a operation if applying that operation to elements of the set always produces an element of the set. A subspace of $\mathbb{R}^n$ is a subset that contains the zero vector, is closed under addition, and is closed under scalar multiplication.

The span of a set of vectors is a subspace. Each nontrivial subspace of $\mathbb{R}^2$ is a line through the origin. Each nontrivial subspace of $\mathbb{R}^3$ is a line or plane through the origin.